

CASEORA

“One Case. One Timeline. Complete Control.”

AI-Powered Advocate Case Management System — Supabase PostgreSQL 16 Database Architecture & Schema Specification

TARGET ENGINE	VECTOR SEARCH	SECURITY MODEL	DATE GENERATED
PostgreSQL 16 (Supabase Cloud)	pgvector (VECTOR 1536, HNSW)	RLS on All 22 Tables	October 2026

1. Executive Summary & Core Architectural Principles

This document defines the formal database architecture for **CASEORA**, a mission-critical legal SaaS tailored for Indian advocates, chambers, and law firms. Grounded in the source Developer Blueprint and operational requirements, the database is architected around 4 immutable pillars:

- 1. Strict AI Case Isolation (Zero Leakage)**

Every AI query, vector chunk, and embedding search enforces strict `WHERE case_id = target_case_id` filtering. Embeddings for Case A can never be retrieved when querying Case B. Context is ephemeral and destroyed upon query completion.
- 2. Absolute Advocate Privilege & Confidentiality**

Indian advocates maintain strict legal professional privilege. The `case_diaries` table is protected via PostgreSQL Row Level Security (RLS) so that **Court Clerks and Clients are strictly blocked** from viewing private trial notes and strategic submissions.
- 3. Automated 2-Slot Limit on Free Tier**

A database trigger on `team_members` automatically intercepts insertions. If a firm is on the `free` tier and already has 2 active team slots, the database raises an exception (`P0001`), enforcing upgrade to the Premium paywall.
- 4. Dynamic Case Health & Readiness (0–100%)**

An automated trigger audits case deadlines. Points are dynamically deducted for overdue tasks (-15 pts) and uncompiled court orders past compliance deadlines (-20 pts), giving advocates instant visibility into readiness.

2. Credentials, Secrets & Integration Mapping

KEY / SECRET	SOURCE	USAGE DESTINATION	SECURITY CLASSIFICATION
<code>SUPABASE_URL</code>	Supabase Dashboard → API	Flutter App & Edge Functions	PUBLIC / APP
<code>SUPABASE_ANON_KEY</code>	Supabase Dashboard → API	Flutter App Initialization	PUBLIC / APP (PROTECTED BY RLS)
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Supabase Dashboard → API	Edge Functions ONLY	CRITICAL (BYPASSES ALL RLS)
<code>GOOGLE_AI_API_KEY</code>	Google AI Studio (Gemini)	Edge Functions (Gemini Pro & Flash)	SECRET (SERVER-SIDE ONLY)
<code>OPENAI_API_KEY</code>	OpenAI Platform	Edge Functions (Vector Embeddings)	SECRET (1536-DIM EMBEDDINGS)
<code>RAZORPAY_KEY_ID + SECRET</code>	Razorpay Dashboard	Edge Functions + Flutter Checkout	SECRET & PUBLIC KEY
<code>google-services.json</code>	Firebase Console	Android App Root	PUSH NOTIFICATION SERVICE

3. PostgreSQL Extensions & Custom Types (Enums)

Required Extensions: `uuid-oss` (UUID generation), `pgcrypto` (Cryptographic hashing), `vector` (pgvector for semantic search).

ENUM NAME	VALUES	DESCRIPTION & LEGAL DOMAIN CONTEXT
<code>user_role</code>	<code>admin_advocate</code> , <code>associate</code> , <code>clerk</code> , <code>client</code>	User authorization levels across the chamber.
<code>subscription_tier</code>	<code>free</code> , <code>premium</code> , <code>enterprise</code>	Law firm billing plan (Free capped at 2 team slots).
<code>case_type</code>	<code>regular_civil_suit</code> , <code>criminal_trial</code> , <code>bail_application</code> , <code>writ_petition</code> , <code>special_leave_petition</code> , <code>cheque_bounce</code> , <code>motor_accident_claim</code> , <code>matrimonial_dispute</code> , <code>insolvency_company</code> , <code>consumer_dispute</code> , <code>arbitration_matter</code> , <code>labor_service</code>	12 Indian Legal Case Categories recognized across District Courts, High Courts, and Tribunals.
<code>folder_category</code>	<code>petition_plaint</code> , <code>written_statements</code> , <code>exhibits</code> , <code>court_orders</code> , <code>interim_applications</code> , <code>affidavits</code> , <code>vakalatnama_memo</code> , <code>evidence_depositions</code> , <code>police_reports</code> , <code>legal_notices</code> , <code>precedents_authorities</code> , <code>voice_memos</code> , <code>correspondence</code> , <code>billing_receipts</code>	14 Case Vault Folders segregating all legal filings, marked exhibits, and audio recordings.
<code>case_status</code>	<code>pre_filing</code> , <code>filing_scrutiny</code> , <code>summons_issued</code> , <code>written_statement_pending</code> , <code>framing_of_issues</code> , <code>plaintiff_evidence</code> , <code>defendant_evidence</code> , <code>final_arguments</code> , <code>reserved_for_orders</code> , <code>disposed</code> , <code>appealed</code>	Statutory procedural stages according to the Code of Civil & Criminal Procedure.
<code>case_priority</code>	<code>urgent</code> , <code>high</code> , <code>medium</code> , <code>routine</code>	Urgency rating (Bail and Injunctions marked Urgent).
<code>advocate_client_role</code>	<code>plaintiff</code> , <code>defendant</code> , <code>petitioner</code> , <code>respondent</code> , <code>appellant</code> , <code>complainant</code> , <code>accused</code> , <code>applicant</code> , <code>other</code>	Representation designation of our client.
<code>disposal_result</code>	<code>allowed</code> , <code>dismissed</code> , <code>partly_allowed</code> , <code>bail_granted</code> , <code>bail_rejected</code> , <code>acquitted</code> , <code>convicted</code> , <code>settled</code> , <code>compromised</code> , <code>withdrawn</code> , <code>transferred</code> , <code>quashed</code> , <code>other</code>	13 Court Disposal Outcomes recorded upon final judgment with appeal limitation dates.
<code>order_type</code>	<code>interim_injunction</code> , <code>status_quo</code> , <code>bail_granted</code> , <code>bail_rejected</code> , <code>adjournment</code> , <code>summons_notice</code> , <code>final_judgment</code> , <code>decree</code> , <code>exemption_order</code> , <code>other</code>	Court order classification.
<code>task_status</code>	<code>pending</code> , <code>in_progress</code> , <code>completed</code> , <code>cancelled</code>	Workflow deadline status.
<code>billing_status</code>	<code>unpaid</code> , <code>partially_paid</code> , <code>paid</code> , <code>overdue</code>	Client invoice settlement state.

4. Complete Data Dictionary — The 22 Core Tables

Module A: Authentication, Chambers & Team Management

1. Table: `firms`

Multi-tenant organization root representing an advocate's chambers or law firm.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Unique firm identifier
<code>firm_name</code>	<code>VARCHAR(255)</code>	<code>NOT NULL</code>	Name of the law firm or chambers
<code>address</code>	<code>TEXT</code>	<code>NULLABLE</code>	Chamber address / court annex
<code>phone_number</code>	<code>VARCHAR(50)</code>	<code>NULLABLE</code>	Contact phone number
<code>email</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Firm correspondence email
<code>admin_advocate_id</code>	<code>UUID</code>	<code>FK → profiles(id)</code>	Managing partner / head counsel
<code>subscription_tier</code>	<code>subscription_tier</code>	<code>NOT NULL, DEFAULT 'free'</code>	Active plan (free / premium / enterprise)
<code>razorpay_customer_id</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Razorpay customer identifier
<code>razorpay_subscription_id</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Active subscription ID
<code>max_team_slots</code>	<code>INT</code>	<code>NOT NULL, DEFAULT 2</code>	Max allowed team seats (2 on free tier)
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

2. Table: `profiles`

User profile linked directly to Supabase Auth (`auth.users`).

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, FK → auth.users(id) ON DELETE CASCADE</code>	Supabase user ID
<code>firm_id</code>	<code>UUID</code>	<code>FK → firms(id) ON DELETE SET NULL</code>	Associated law firm
<code>role</code>	<code>user_role</code>	<code>NOT NULL, DEFAULT 'admin_advocate'</code>	Role (admin_advocate, associate, clerk, client)
<code>full_name</code>	<code>VARCHAR(255)</code>	<code>NOT NULL</code>	Advocate / user full name
<code>phone_number</code>	<code>VARCHAR(50)</code>	<code>NULLABLE</code>	Mobile contact
<code>enrollment_number</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Bar Council Enrollment No (e.g. MAH/1234/2012)
<code>bar_council_state</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	State Bar Council (e.g. Maharashtra & Goa)
<code>office_address</code>	<code>TEXT</code>	<code>NULLABLE</code>	Chamber office address
<code>specialization</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Civil, Criminal, Constitutional, NCLT, etc.
<code>profile_photo_url</code>	<code>TEXT</code>	<code>NULLABLE</code>	Avatar image URL
<code>is_biometric_enabled</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT false</code>	Biometric app lock toggle

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

3. Table: `team_members`

Tracks chamber associate and clerk assignments with trigger-enforced 2-member limit.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Unique slot ID
<code>firm_id</code>	<code>UUID</code>	<code>FK → firms(id) ON DELETE CASCADE</code>	Firm ID
<code>user_id</code>	<code>UUID</code>	<code>FK → profiles(id) ON DELETE CASCADE</code>	Assigned team member profile
<code>role</code>	<code>user_role</code>	<code>NOT NULL, DEFAULT 'associate'</code>	Team member role
<code>is_active</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT true</code>	Seat active status
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

Module B: Litigants & Case Engine

4. Table: `clients`

Litigant records managed by the chambers.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Unique client ID
<code>firm_id</code>	<code>UUID</code>	<code>FK → firms(id) ON DELETE CASCADE</code>	Firm ID
<code>client_name</code>	<code>VARCHAR(255)</code>	<code>NOT NULL</code>	Full name of litigant
<code>phone_number</code>	<code>VARCHAR(50)</code>	<code>NOT NULL</code>	Contact phone number
<code>email</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Litigant email
<code>address</code>	<code>TEXT</code>	<code>NULLABLE</code>	Residential / commercial address
<code>portal_access_enabled</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT false</code>	Can litigant log into Client Portal
<code>portal_user_id</code>	<code>UUID</code>	<code>FK → auth.users(id) ON DELETE SET NULL</code>	Linked login user ID
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

5. Table: `cases`

The core master entity for all legal matters handled by the firm.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Primary case identifier
<code>firm_id</code>	<code>UUID</code>	<code>FK → firms(id) ON DELETE CASCADE</code>	Firm ID
<code>case_number</code>	<code>VARCHAR(100)</code>	<code>NOT NULL</code>	Official Court Case Number (e.g. RCS/412/2026)
<code>filing_number</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Initial filing lodging number
<code>registration_number</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Court registration number
<code>case_type</code>	<code>case_type</code>	<code>NOT NULL</code>	One of the 12 Indian legal case types
<code>year</code>	<code>INT</code>	<code>NOT NULL</code>	Filing year
<code>filing_date</code>	<code>DATE</code>	<code>NULLABLE</code>	Date of plaint/petition filing
<code>registration_date</code>	<code>DATE</code>	<code>NULLABLE</code>	Date registered by court registry
<code>case_status</code>	<code>case_status</code>	<code>NOT NULL, DEFAULT 'pre_filing'</code>	Procedural status
<code>case_priority</code>	<code>case_priority</code>	<code>NOT NULL, DEFAULT 'medium'</code>	Priority rating (urgent / high / medium / routine)
<code>client_id</code>	<code>UUID</code>	<code>FK → clients(id) ON DELETE RESTRICT</code>	Represented client ID
<code>advocate_client_role</code>	<code>advocate_client_role</code>	<code>NOT NULL, DEFAULT 'petitioner'</code>	Client's designation (Plaintiff, Accused, etc.)
<code>lead_advocate_id</code>	<code>UUID</code>	<code>FK → profiles(id) ON DELETE RESTRICT</code>	Lead counsel handling the brief
<code>court_name</code>	<code>VARCHAR(255)</code>	<code>NOT NULL</code>	Court name (e.g. City Civil Court, Dindoshi)
<code>court_number</code>	<code>VARCHAR(50)</code>	<code>NULLABLE</code>	Courtroom / Hall number
<code>judge_name</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Presiding Judicial Officer / Bench
<code>court_address</code>	<code>TEXT</code>	<code>NULLABLE</code>	Court complex physical address
<code>district / state</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	Jurisdiction district and state
<code>cnr_number</code>	<code>VARCHAR(50)</code>	<code>NULLABLE</code>	16-digit eCourts CNR Number
<code>fir_number</code>	<code>VARCHAR(100)</code>	<code>NULLABLE</code>	FIR Number (if criminal/police matter)
<code>police_station</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Jurisdictional police station
<code>relevant_acts_sections</code>	<code>TEXT</code>	<code>NULLABLE</code>	Statutory sections (e.g. Sec 138 NI Act, Sec 38 SRA)
<code>related_cases</code>	<code>TEXT</code>	<code>NULLABLE</code>	Cross-referenced connected matters
<code>health_score</code>	<code>INT</code>	<code>NOT NULL, DEFAULT 100, CHECK (0-100)</code>	Dynamic case readiness health score (0–100%)
<code>is_disposed</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT false</code>	Case disposed / closed flag

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

6. Table: `case_assignments` & `case_parties`

Junction table linking multiple associates to cases, and registry of all litigants & opposing counsels.

TABLE	KEY COLUMNS	CONSTRAINTS	PURPOSE
<code>case_assignments</code>	<code>case_id, user_id, assigned_at</code>	<code>FK → cases, FK → profiles, UNIQUE(case_id, user_id)</code>	Many-to-many associate assignments for RLS scoping.
<code>case_parties</code>	<code>case_id, party_name, party_designation, contact_details, is_our_party, advocate_name, advocate_contact</code>	<code>FK → cases(id) ON DELETE CASCADE</code>	Stores all parties (Plaintiff, Defendant No. 1, etc.) and opposing counsel contact details.

Module C: Hearings, Private Diary & Exemptions

7. Table: `case_hearings`

Scheduled court hearings, cause lists, and proceedings log.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Hearing ID
<code>case_id</code>	<code>UUID</code>	<code>FK → cases(id) ON DELETE CASCADE</code>	Linked case
<code>court_name</code>	<code>VARCHAR(255)</code>	<code>NULLABLE</code>	Courtroom / Bench
<code>hearing_date</code>	<code>DATE</code>	<code>NOT NULL</code>	Scheduled hearing date
<code>purpose</code>	<code>TEXT</code>	<code>NOT NULL</code>	Purpose of hearing (e.g. Framing of Issues)
<code>previous_proceedings</code>	<code>TEXT</code>	<code>NULLABLE</code>	Summary of what transpired previously
<code>next_action</code>	<code>TEXT</code>	<code>NULLABLE</code>	Action required before this hearing
<code>next_hearing_date</code>	<code>DATE</code>	<code>NULLABLE</code>	Next date fixed by the court
<code>is_attended</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT false</code>	Appearance marked
<code>reminder_sent</code>	<code>BOOLEAN</code>	<code>NOT NULL, DEFAULT false</code>	Push notification reminder status
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

8. Table: `case_diaries` **STRICTLY PRIVILEGED**

Confidential advocate trial diary. Protected by RLS: **Clerks and Clients are prohibited.**

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Diary entry ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>hearing_date</code>	DATE	NOT NULL	Date of hearing
<code>what_happened</code>	TEXT	NOT NULL	Detailed account of courtroom events
<code>court_remarks</code>	TEXT	NULLABLE	Observations made by the presiding bench
<code>advocate_submissions</code>	TEXT	NULLABLE	Legal arguments submitted by counsel
<code>is_client_present</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Litigant presence flag
<code>is_opposite_present</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Opposite party presence flag
<code>is_exemption_filed</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Was absence exemption application filed
<code>court_decision</code>	TEXT	NOT NULL	Order / direction passed by the court
<code>next_steps</code>	TEXT	NULLABLE	Advocate's confidential strategic action points
<code>audio_memo_url</code>	TEXT	NULLABLE	Supabase Storage path for recorded voice memo
<code>raw_transcript</code>	TEXT	NULLABLE	Multilingual transcript (English / Marathi / Hindi)
<code>recorded_by</code>	UUID	FK → <code>profiles(id)</code> ON DELETE RESTRICT	Advocate who authored the diary
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

9. Table: `exemption_applications`

Tracks litigant absence exemption applications, reasons, court responses, and costs imposed.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Exemption ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>hearing_date</code>	DATE	NOT NULL	Hearing date of absence
<code>client_name</code>	VARCHAR(255)	NOT NULL	Name of absent client
<code>reason_for_absence</code>	TEXT	NOT NULL	Illness, out of town, emergency, etc.
<code>supporting_document_url</code>	TEXT	NULLABLE	Medical certificate / flight ticket URL
<code>submission_date</code>	DATE	NOT NULL, DEFAULT <code>CURRENT_DATE</code>	Date application filed in court
<code>court_response</code>	TEXT	NULLABLE	Allowed / Rejected / Warning
<code>order_text</code>	TEXT	NULLABLE	Formal court exemption order
<code>cost_imposed</code>	NUMERIC(10, 2)	DEFAULT <code>0.00</code>	Court costs / library fund penalty (INR)
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

Module D: Tasks, Document Vault, Exhibits & Orders

10. Table: `case_tasks`

Statutory deadlines and team to-dos (Prepare reply, file exemption, bail application, etc.).

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Task ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>title</code>	VARCHAR(255)	NOT NULL	Task title (e.g. Prepare Reply to Injunction)
<code>description</code>	TEXT	NULLABLE	Detailed instructions
<code>due_date</code>	TIMESTAMPTZ	NOT NULL	Statutory deadline
<code>priority</code>	<code>task_priority</code>	NOT NULL, DEFAULT 'medium'	urgent / high / medium / low
<code>status</code>	<code>task_status</code>	NOT NULL, DEFAULT 'pending'	pending / in_progress / completed / cancelled
<code>assigned_to</code>	UUID	FK → <code>profiles(id)</code> ON DELETE SET NULL	Assigned associate or clerk
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

11. Table: `case_documents`

The 14-Folder Document Vault. Supports PDF, DOCX, JPG, PNG, and scanned legal documents.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Document ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>firm_id</code>	UUID	FK → <code>firms(id)</code> ON DELETE CASCADE	Firm ID
<code>category</code>	<code>folder_category</code>	NOT NULL	One of the 14 defined folder categories
<code>filename</code>	VARCHAR(255)	NOT NULL	Original file name
<code>file_ext</code>	VARCHAR(20)	NOT NULL	File extension (pdf, docx, jpg, png)
<code>file_size_bytes</code>	BIGINT	NOT NULL	File size in bytes
<code>storage_path</code>	TEXT	NOT NULL	Supabase Storage URI (<code>case-vault/{firm}/{case}/{folder}/{file}</code>)
<code>ocr_completed</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Has OCR pipeline completed text extraction
<code>extracted_text</code>	TEXT	NULLABLE	Full OCR text content
<code>ai_summary</code>	TEXT	NULLABLE	AI document summary
<code>is_client_accessible</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Can litigant see this file in Client Portal
<code>uploaded_by</code>	UUID	FK → <code>profiles(id)</code> ON DELETE RESTRICT	Advocate who uploaded the file
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

12. Table: `case_exhibits`

Formally marked trial exhibits (Exh. P-1 to P-n, Exh. D-1 to D-n) with witness linking.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Exhibit ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>exhibit_number</code>	VARCHAR(50)	NOT NULL	Formal marked number (e.g. Exh. P-1, Exh. D-4)
<code>document_id</code>	UUID	FK → <code>case_documents(id)</code> ON DELETE SET NULL	Linked document in vault
<code>document_name</code>	VARCHAR(255)	NOT NULL	Name of marked document
<code>description</code>	TEXT	NULLABLE	Description of evidence
<code>witness_name</code>	VARCHAR(255)	NULLABLE	Witness who deposited/proved the exhibit
<code>upload_date</code>	DATE	NOT NULL, DEFAULT <code>CURRENT_DATE</code>	Registry date
<code>related_hearing_id</code>	UUID	FK → <code>case_hearings(id)</code> ON DELETE SET NULL	Hearing at which exhibit was marked
<code>notes</code>	TEXT	NULLABLE	Advocate notes regarding admissibility/objection
<code>marking_date</code>	DATE	NULLABLE	Date marked by judge
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

13. Table: `case_orders`

Judicial orders, interim injunctions, bail conditions, and compliance deadlines.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Order record ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>order_date</code>	DATE	NOT NULL	Date order pronounced
<code>order_type</code>	<code>order_type</code>	NOT NULL	Interim Injunction, Bail Granted, Stay, etc.
<code>document_id</code>	UUID	FK → <code>case_documents(id)</code> ON DELETE SET NULL	Original signed order PDF in vault
<code>raw_order_text</code>	TEXT	NULLABLE	Verbatim operative part of the order
<code>advocate_notes</code>	TEXT	NULLABLE	Advocate commentary & conditions imposed
<code>ai_order_summary</code>	TEXT	NULLABLE	AI digest of terms, conditions, & deadlines
<code>compliance_required</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Does order mandate filing compliance report
<code>compliance_deadline</code>	TIMESTAMPTZ	NULLABLE	Compliance deadline (triggers alert)
<code>is_complied</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Compliance satisfied flag

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>created_at</code> / <code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Audit timestamps

Module E: Case AI Brain & Semantic Search (pgvector)

14. Table: `case_ai_brain_state` 1-TO-1 WITH CASES

Persistent synthesis state of the isolated Case AI Brain.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>case_id</code>	<code>UUID</code>	<code>PK, FK → cases(id) ON DELETE CASCADE</code>	1-to-1 linkage with case
<code>facts_summary</code>	<code>TEXT</code>	<code>NULLABLE</code>	AI summary of core material facts
<code>what_happened_summary</code>	<code>TEXT</code>	<code>NULLABLE</code>	Chronological digest of past hearings
<code>what_next_actions</code>	<code>JSONB</code>	<code>DEFAULT '[]'::jsonb</code>	Array of predicted next procedural actions
<code>readiness_factors</code>	<code>JSONB</code>	<code>DEFAULT '{}'::jsonb</code>	Checklist of missing pleadings, evidence, fees
<code>legal_strategy</code>	<code>JSONB</code>	<code>DEFAULT '{}'::jsonb</code>	Recommended legal arguments & defenses
<code>predicted_outcomes</code>	<code>JSONB</code>	<code>DEFAULT '{}'::jsonb</code>	Outcome probability analysis & risks
<code>updated_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Timestamp of last AI synthesis

15. Table: `case_document_embeddings` ISOLATED VECTOR STORE

Vector chunks for semantic retrieval. Enforces `case_id` partitioning to guarantee zero data leakage.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Vector chunk ID
<code>case_id</code>	<code>UUID</code>	<code>FK → cases(id) ON DELETE CASCADE</code>	MANDATORY ISOLATION PARTITION KEY
<code>document_id</code>	<code>UUID</code>	<code>FK → case_documents(id) ON DELETE CASCADE</code>	Source document in vault
<code>content_chunk</code>	<code>TEXT</code>	<code>NOT NULL</code>	Text content of the 500-1000 token chunk
<code>chunk_index</code>	<code>INT</code>	<code>NOT NULL</code>	Sequential chunk position within document
<code>embedding</code>	<code>VECTOR(1536)</code>	<code>NOT NULL</code>	1536-dimensional vector (OpenAI / Gemini)
<code>created_at</code>	<code>TIMESTAMPTZ</code>	<code>NOT NULL, DEFAULT now()</code>	Timestamp

16. Table: `legal_precedents`

Cached Indian Precedent Library (Supreme Court & High Courts) with semantic search.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	<code>UUID</code>	<code>PK, DEFAULT uuid_generate_v4()</code>	Precedent ID
<code>citation</code>	<code>VARCHAR(255)</code>	<code>NOT NULL, UNIQUE</code>	Legal Citation (e.g. 1994 SCC (4) 225)
<code>case_title</code>	<code>VARCHAR(255)</code>	<code>NOT NULL</code>	Petitioner v. Respondent
<code>court_name</code>	<code>VARCHAR(100)</code>	<code>NOT NULL</code>	Supreme Court of India, Bombay HC, etc.
<code>decision_year</code>	<code>INT</code>	<code>NULLABLE</code>	Year of judgment
<code>relevant_acts_sections</code>	<code>TEXT</code>	<code>NULLABLE</code>	Acts interpreted (e.g. Order 39 Rule 3 CPC)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
ratio_decidendi	TEXT	NOT NULL	Core legal principle / binding proposition
full_judgment_url	TEXT	NULLABLE	Link to complete judgment text
embedding	VECTOR(1536)	NULLABLE	Vector representation of the ratio
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT now()	Timestamp

Module F: Timeline, Disposal, Financials, Messages & Audit

17. Table: `case_timeline_events`

Sequential visual timeline stages (Filing → Summons → WS → Evidence → Arguments → Judgment).

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	UUID	PK, DEFAULT <code>uuid_generate_v4()</code>	Timeline event ID
<code>case_id</code>	UUID	FK → <code>cases(id)</code> ON DELETE CASCADE	Linked case
<code>stage_name</code>	VARCHAR(100)	NOT NULL	Stage name (e.g. Plaintiff Filed, Summons Served)
<code>description</code>	TEXT	NULLABLE	Stage narrative
<code>event_date</code>	DATE	NULLABLE	Date stage reached / completed
<code>is_completed</code>	BOOLEAN	NOT NULL, DEFAULT <code>false</code>	Completed toggle
<code>order_index</code>	INT	NOT NULL, DEFAULT 1	Display order in stepper UI
<code>related_document_id</code>	UUID	FK → <code>case_documents(id)</code> ON DELETE SET NULL	Document proving stage completion
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

18. Table: `case_disposals`

Final case resolution records, judgment links, and statutory appeal limitation deadlines.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>case_id</code>	UUID	PK, FK → <code>cases(id)</code> ON DELETE CASCADE	1-to-1 linkage with disposed case
<code>disposal_date</code>	DATE	NOT NULL	Date of final decree / judgment
<code>final_result</code>	<code>disposal_result</code>	NOT NULL	Allowed, Dismissed, Acquitted, Convicted, etc.
<code>judgment_order_url</code>	TEXT	NULLABLE	Storage URL of final judgment copy
<code>disposal_reason</code>	TEXT	NULLABLE	Grounds of disposal
<code>final_documents</code>	JSONB	DEFAULT <code>'[]'::jsonb</code>	List of final orders & decree sheets
<code>advocate_notes</code>	TEXT	NULLABLE	Advice on appeal / execution
<code>appeal_deadline</code>	DATE	NULLABLE	Statutory limitation expiry for appeal (e.g. 90 days)
<code>created_at</code> / <code>updated_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT <code>now()</code>	Audit timestamps

19. Table: `billings` & `case_expenses`

Fee invoicing, auto-calculated balances, court expenses, and travel reimbursements.

TABLE	KEY COLUMNS	CONSTRAINTS & COMPUTATIONS	DESCRIPTION
billings	case_id, client_id, invoice_number, total_fee, paid_amount, pending_amount, invoice_items, status, due_date, invoice_pdf_url	invoice_number UNIQUE pending_amount GENERATED ALWAYS AS (total_fee - paid_amount) STORED	Client fee invoices. Automatically calculates pending balance. Generates invoice PDFs.
case_expenses	case_id, expense_name, expense_category, amount, receipt_url, is_reimbursed, logged_by	FK → cases(id), FK → profiles(id)	Logs out-of-pocket court fees, typing, process fee, and advocate travel with receipt uploads.

20. Table: client_messages & audit_logs

Encrypted in-app advocate ↔ client communications and immutable firm audit trails.

TABLE	KEY COLUMNS	CONSTRAINTS	DESCRIPTION
client_messages	case_id, sender_id, recipient_id, message_text, attachment_url, is_read, created_at	FK → cases(id), FK → auth.users(id)	Secure case-linked communication channel between advocate and litigant.
audit_logs	firm_id, user_id, case_id, action_type, metadata, ip_address, user_agent, created_at	FK → firms(id), FK → auth.users(id)	Immutable trail of actions (e.g. DIARY_ACCESSED, EXPORT_PDF, ORDER_CREATED). Accessible only to Admin Advocate.

5. Database Triggers & Business Logic Stored Procedures

Trigger 1: Enforce Free Plan 2-Member Limit

Blocks insertion of a 3rd team member when a firm is on the Free tier.

```
CREATE OR REPLACE FUNCTION trg_enforce_team_slot_limit()
RETURNS TRIGGER AS $$
DECLARE
    current_tier subscription_tier;
    member_count INT;
    allowed_slots INT;
BEGIN
    SELECT subscription_tier, max_team_slots
    INTO current_tier, allowed_slots
    FROM firms WHERE id = NEW.firm_id;

    IF current_tier = 'free' THEN
        SELECT COUNT(*) INTO member_count
        FROM team_members
        WHERE firm_id = NEW.firm_id AND is_active = true;

        IF member_count >= allowed_slots THEN
            RAISE EXCEPTION 'Free plan allows a maximum of % team members. Upgrade to Premium to add more members.', allowed_s
            USING ERRCODE = 'P0001';
        END IF;
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER check_team_slot_limit_before_insert
BEFORE INSERT ON team_members
FOR EACH ROW EXECUTE FUNCTION trg_enforce_team_slot_limit();
```

Trigger 2: Dynamic Case Health Score Recalculator (0–100%)

Monitors task deadlines and court order compliance, adjusting the case health score in real-time.

```

CREATE OR REPLACE FUNCTION trg_update_case_health_score()
RETURNS TRIGGER AS $$
DECLARE
    target_case_id UUID;
    score INT := 100;
    overdue_tasks_count INT := 0;
    pending_orders_count INT := 0;
BEGIN
    target_case_id := COALESCE(NEW.case_id, OLD.case_id);

    -- Deduct 15 points per overdue task
    SELECT COUNT(*) INTO overdue_tasks_count
    FROM case_tasks
    WHERE case_id = target_case_id
        AND status != 'completed'
        AND due_date < timezone('utc', now());
    score := score - (overdue_tasks_count * 15);

    -- Deduct 20 points per uncomplied court order past deadline
    SELECT COUNT(*) INTO pending_orders_count
    FROM case_orders
    WHERE case_id = target_case_id
        AND compliance_required = true
        AND is_complied = false
        AND compliance_deadline < timezone('utc', now());
    score := score - (pending_orders_count * 20);

    IF score < 0 THEN score := 0; END IF;

    UPDATE cases
    SET health_score = score, updated_at = timezone('utc', now())
    WHERE id = target_case_id;

    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

```

6. Vector Similarity Search RPC Functions (Isolated pgvector)

```

-- 1. Search Case Documents with MANDATORY Case Isolation
CREATE OR REPLACE FUNCTION match_case_documents (
    target_case_id UUID,
    query_embedding VECTOR(1536),
    match_count INT DEFAULT 5,
    match_threshold FLOAT DEFAULT 0.65
)
RETURNS TABLE (
    id UUID,
    document_id UUID,
    content_chunk TEXT,
    similarity FLOAT
)
LANGUAGE plpgsql SECURITY DEFINER AS $$
BEGIN
    RETURN QUERY
    SELECT
        cde.id,
        cde.document_id,
        cde.content_chunk,
        1 - (cde.embedding <=> query_embedding) AS similarity
    FROM case_document_embeddings cde
    WHERE cde.case_id = target_case_id -- STRICT ISOLATION FILTER
        AND 1 - (cde.embedding <=> query_embedding) > match_threshold
    ORDER BY cde.embedding <=> query_embedding
    LIMIT match_count;
END;
$$;

```

7. Row Level Security (RLS) Policy Access Matrix

TABLE / RESOURCE	ADMIN ADVOCATE	ASSOCIATE ADVOCATE	COURT CLERK	CLIENT (LITIGANT)
firms & profiles	Full Access	View Chamber Members	View Chamber Members	View Own Profile
cases	All firm cases	Assigned cases only	Read-only (all firm cases)	Own registered cases only
case_diaries	Full CRUD	Assigned cases only	STRICTLY BLOCKED	STRICTLY BLOCKED
case_documents	Full CRUD	Assigned cases only	Read-only	is_client_accessible = true only
case_exhibits & orders	Full CRUD	Assigned cases only	Read-only	Read-only (their cases)
case_ai_brain_state	Full CRUD	Assigned cases only	STRICTLY BLOCKED	STRICTLY BLOCKED
case_document_embeddings	Full CRUD	Assigned cases only	STRICTLY BLOCKED	STRICTLY BLOCKED
billings	Full CRUD	Read-only	Read-only	Own invoices only
audit_logs	Read (own firm)	STRICTLY BLOCKED	STRICTLY BLOCKED	STRICTLY BLOCKED

8. Storage Bucket Architecture (case-vault)

Bucket Name: case-vault (Private, Non-Public)

Hierarchical Path Convention:

case-vault/{firm_id}/{case_id}/{folder_category}/{filename}.pdf

Real Examples:

- case-vault/firm_01/case_1092/petition_plaint/Civil_Suit_Plaint.pdf
- case-vault/firm_01/case_1092/exhibits/exh_p1_agreement.pdf
- case-vault/firm_01/case_1092/voice_memos/hearing_2026-10-02.m4a

Storage Policies: Upload and delete permitted only to authenticated advocates matching their firm_id .